

SQL

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1. Write the following inserts, deletes or updates in SQL, using the university schema.
 - a. Increase the salary of each instructor in the Comp. Sci. department by 10%.
 - b. Delete all courses that have never been offered (that is, do not occur in the section relation).
 - c. Insert every student whose tot_cred attribute is greater than 100 as an instructor in the same department, with a salary of \$10,000.
 - d. create a new course "CS-001", titled with "Weekly Seminar", with 0 credits.
 - e. Create a section of this course in Fall 2009, with sec_id of 1.
 - f. Enroll every student in the Comp. Sci. department in the above section.
 - g. Delete enrollments in the above section where the student's name is Chavez.
 - h. Delete the course CS-101. What will happen if you run this delete statement without first deleting offerings (sections) of this course.
 - i. Delete all takes tuples corresponding to any section of any course with the word "database" as a part of the title; ignore case when matching the word with the title.

2. Consider the following employee database

```
employee(employee_name, street, city)
works(employee_name, company_name, salary)
company(company_name, city)
manages(employee_name, manager_name)
```

Give an expression in SQL for each of the following queries.

- a. Find the names and cities of residence of all employees who work for "First Bank Corporation".
- b. Find the names, street addresses, and cities of residence of all employee who works for "First Bank Corporation" and earn more than \$10,000.
- c. Find all employees in the database who do not work for "First Bank Corporation".
- d. Find all employees in the database who earn more than each employee of "Small Bank Corporation".
- e. Assume that the companies may be located in several cities. Find all companies located in every city in which "Small Bank Corporation" is located.
- f. Find the company that has the most employees.
- g. Find those companies whose employees earn a higher salary, on average, than the average salary at "First Bank Corporation".